cent, since the male had no father and obtained all his genes from his mother. But now suppose a queen is known to have the gene *B*. The chance that her son shares the gene is only 50 per cent, since he contains only half her genes. This sounds like a contradiction, but it is not. A male gets *all* his genes from his mother, but a mother only gives *half* her genes to her son. The solution to the apparent paradox lies in the fact that a male has only half the usual number of genes. There is no point in puzzling over whether the 'true' index of relatedness is $\frac{1}{2}$ or 1. The index is only a man-made measure, and if it leads to difficulties in particular cases, we may have to abandon it and go back to first principles. From the point of view of a gene *A* in the body of a queen, the chance that the gene is shared by a son is $\frac{1}{2}$, just as it is for a daughter. From a queen's point of view therefore, her offspring, of either sex, are as closely related to her as human children are to their mother.

Things start to get intriguing when we come to sisters. Full sisters not only share the same father: the two sperms that conceived them were identical in every gene. The sisters are therefore equivalent to identical twins as far as their paternal genes are concerned. If one female has a gene *A*, she must have got it from either her father or her mother. If she got it from her mother then there is a 50 per cent chance that her sister shares it. But if she got it from her father, the chances are 100 per cent that her sister shares it. Therefore the relatedness between hymenopteran full sisters is not $\frac{1}{2}$ as it would be for normal sexual animals, but $\frac{3}{4}$.

It follows that a hymenopteran female is more closely related to her full sisters than she is to her offspring of either sex.* As Hamilton realized (though he did not put it in quite the same way) this might well predispose a female to farm her own mother as an efficient sister-making machine. A gene for vicariously making sisters replicates itself more rapidly than a gene for making offspring directly. Hence worker sterility evolved. It is presumably no accident that true sociality, with worker sterility, seems to have evolved no fewer than eleven times *independently* in the Hymenoptera and only once in the whole of the rest of the animal kingdom, namely in the termites.

However, there is a catch. If the workers are successfully to farm their mother as a sister-producing machine, they must somehow curb her